

Answers

1. One function of a modern PCs operating system is multi-tasking. This allows two or more programs to be open at the same time.

(a) Describe how multi-tasking works. [4]

When a program is opened, the program instructions are copied into RAM (1) along with any data/files that the program needs (1). Other programs can also be opened and copied into RAM (1). The operating system manages for one of the programs to get a small amount of CPU time (1). When it has this time, the instructions are executed (1). The CPU will then make the program wait whilst another program gets CPU time (1). As this happens very quickly, the user will experience one or more programs appearing to run at the same time (1) even if there is only one CPU (1).

(b) When a user presses a key on their keyboard an interrupt is generated. Describe the actions of the operating system as a result of this interrupt. [2]

The interrupt is handled by the interrupt service routine / ISR / interrupt handler. (1)
When the interrupt signal is received, the CPU will pause its current task (1) and then run the ISR (1).

Once the ISR has been run, the CPU will return to running the previous task it was on / next task in the queue (1).

(c) A keypress on a keyboard is an example of a hardware interrupt.

Give **one** example of a software interrupt. [1]

Two processes trying to access the same memory location. (1)
Division by zero. (1)

Accept other examples that will create a software interrupt.

(d) Other than managing multi-tasking and interrupts, explain **three** other functions of an operating system [6]

Any three of...

- Managing files / file management (1) allows for files to be created / deleted / sub-folders / renaming of files (1).
- Provides an interface (1) for the user and computer to communicate with one another (1).
- Manages peripherals / drivers (1) so that additional hardware, such as storage / webcam / printer can be used (1).
- Manages memory / memory management (1) which prevents memory from accessing locations used by other programs / which moves programs and data between RAM and virtual memory as required (1).
- Provides a platform for running applications (1). This means that applications are able to make use of different libraries/function calls that the operating system provides (1).
- Provides system security (1) such as usernames and passwords / restricted access to files / antivirus software / anti-malware software. (1)
Accept any individual security example with an explanation of what it does.
- Manages user accounts (1) so that certain users are able to access certain files/services / by allowing users to be created, deleted or privileges altered (1).

Assessment test

Unit 6 Software

2. A disk defragmenter is an example of a utility program.

(a) Explain what a utility program is. [2]

Systems software that provides additional functionality to the operating system (1) such as housekeeping tasks (1) including formatters / clipboard / disk compression software (1) (allow other examples of utility programs).

(b) A disk defragmenter may improve the performance of a computer.

Describe how a disk defragmenter works to improve the performance of a computer. [3]

- When the hard disk saves a file, it saves it to free space which may not be large enough for the whole file (1).
- This results in parts of the file being in a different location (1).
- Fragments take more time to access (1) (as the drive head needs to move to another track and wait for the sector to rotate under it).
- Defragmentation software will place the fragments of the file together (1).
- Files can be accessed as one contiguous file (1).

(c) Hard drives normally use magnetic or solid-state technologies.

Explain which technology will need defragmentation. [2]

Magnetic (1) hard disks need defragmentation. This is because they need to move a drive head / rotate a disk (1).

3. Zach buys a new computer with an operating system installed. When he first turns it on, the computer runs a bootloader.

(a) The bootloader is firmware. Explain what the term firmware means. [2]

It is program instructions/software (1) that are stored in ROM / hardware / EEPROM / Flash ROM (1).

(b) State the purpose of a bootloader. [1]

It is the program that boots the computer by running the initial instructions required to start the operating system loading / initialise hardware settings (1).

It is responsible for loading the BIOS (1).

It is responsible for loading the boot sector on the hard drive / secondary storage (1).

4. A programmer is creating a device driver for a new webcam.
They make use of assembly language to produce the device driver.
- (a) Identify whether assembly language is a high-level or low-level language. [2]
Low-level language (1).
- (b) The assembly code produced by the programmer makes use of mnemonics.
Give the purpose of using mnemonics when creating assembly code. [1]
The mnemonic represents the operation of an instruction (1). For example, MOV means move a value from memory to a register (1).
Accept other examples of how a mnemonic represents a CPU instruction.
- (c) Give **one** reason why the programmer has chosen assembly language rather than a high-level language to write the device driver. [1]
They are able to directly control the hardware (1).
They can make the device driver more efficient in its use of memory / processing requirements (1).
- (d) Once the programmer has completed their assembly program, state the translator they will need to use to create the machine code. [1]
Assembler. (1)

The programmer is also making an application that will allow users to edit videos made with their webcam.

- (e) Explain why a high-level language would be more appropriate than assembly code for programming this application. [2]
It is easier to read / write / debug code (1).
The software could be compiled to different computer architectures / the software will have machine independence (1).
- (f) Once the video editing application is complete the programmer makes use of a compiler. Describe the purpose of a compiler in this situation. [2]
The compiler is a translator (1) which converts the high-level source code (1) into machine code (1) so that it can be executed/run by the computer (1).
- (g) The programmer could have made use of a high-level language that would require an interpreter rather than a compiler. [1]
State the meaning of the term interpreter.
It translates and executes the program line by line. (1)
- (h) Explain why an interpreter wouldn't be as good a choice as a compiler for the video editing application. [2]
The interpreter will be slower (1) as it will need to compile and run each line of code, (rather than just run it) (1).
If an interpreter was used, each customer would need to install the interpreter in order to be able to run the software (1) which would make the installation process longer / require more RAM / require more storage space (1).
The programmer would need to provide the user with the source code (1) which would mean that the user could see how it works / view it / alter it / steal ideas used in it (1).

5. Most programmers make use of IDE when writing program code.

(a) One key feature of an IDE is a code editor.

Describe **two** features that are available to a programmer that will change how the code appears in the code editor. [4]

Prettyprint (1) will change the indentation / whitespace / formatting of the code (1).

Auto-completion (1) will give a list of options for keywords/variables/functions/procedures/methods/properties that the programmer can select / will complete sections of code automatically (1).

Auto-correction (1) keywords/variable names that are mistyped can be corrected automatically / missing braces/brackets can be automatically added (1).

Syntax highlighting (1) will change the colour/formatting of keywords/variables/strings (to make different types of syntax easier to identify) (1).

(b) Name **one** other feature of an IDE that will help the programmer with debugging their code. [1]

Error diagnostics (1)

Accept:

Breakpoints (1)

Stepping through code (1)

Watching variables (1)

Line numbers to identify the location of syntax errors (1)

[Total 40 marks]